

Project Initialization and Planning Phase

Date	26 June 2026
Team ID	SWTID-2026-9097
Project Name	Voice Based Concept Understanding Analyzer
Maximum Marks	3 Marks

Define Problem Statements (Customer Problem Statement Template):

Assessing conceptual understanding through traditional methods is often time-consuming, subjective, and limited to written evaluations. Students frequently lack instant, personalized feedback on their spoken explanations, making it difficult to identify gaps in understanding and improve communication skills. Educators also face challenges in evaluating multiple students consistently and efficiently. The Voice-Based Concept Understanding Analyzer addresses this problem by using AI-powered speech transcription, semantic analysis, and automated scoring to evaluate spoken responses, provide detailed feedback, and generate comprehensive reports, making concept assessment more accurate, efficient, and personalized.

Problem Statement

Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel

PS-1	A student, educator, or educational institution seeking an effective way to assess conceptual understanding Evaluate and improve spoken conceptual explanations while tracking learning progress through meaningful feedback Frustrated, uncertain about the accuracy of the evaluation, and less confident in making informed learning and teaching decisions.	Evaluate and improve spoken conceptual explanations while tracking learning progress through meaningful feedback Frustrated, uncertain about the accuracy of the evaluation, and less confident in making informed learning and teaching decisions.	Existing assessment methods are time-consuming, inconsistent, and provide limited insights into conceptual understanding, fluency, and communication skills.	Most available tools focus on text-based evaluation or manual assessment and lack AI-powered voice analysis with automated reporting	Frustrated, uncertain about the accuracy of the evaluation, and less confident in making informed learning and teaching decisions.
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